

- Business Requirements Document (BRD)**
- Project Title:** Data Integration and Reporting for Customer Order Data
- Introduction:**

The purpose of this project is to design and implement a data integration and reporting solution that consolidates customer order data from CSV files into a structured database, enabling efficient data analysis and reporting.

- Business Requirements:**

1. **Data Ingestion**

- Read source CSV data from the volume and load it into the Delta tables "customer" and "order".
- Customer data is located at: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata``
- Order data is located at: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``

2. **Data Transformation**

- The schema of the "customer" table is: ``CustId`, `Name`, `EmailId`, `Region``
- The schema of the "order" table is: ``OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId``
- Remove "Null"/null and duplicate records from both tables.

3. **Data Integration**

- Create the "ordersummary" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- The schema of the "ordersummary" table is: ``CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date``
- Join the "customer" and "order" data using the "CustId" field and load the data into an SCD Type 2 table "ordersummary".

4. **SCD Type 2 Implementation**

- Include logic to update the SCD Type 2 "ordersummary" table whenever there is a change in the "customer" table.
- In the "ordersummary" table, old records should be made inactive, and new records should be made active.
- Update the ``StartDate`` and ``EndDate`` accordingly.

5. **Data Aggregation**

- Create a new table "customeraggregatespend" with columns "Name", "TotalAmount", and "Date" in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist.
- Aggregate the "TotalAmount" column from the "ordersummary" table and group it by the "Name" and "Date" columns.

- Load the aggregated data into the "customeraggregatespend" table.

- Functional Requirements:**

1. The system should be able to read CSV data from the specified locations and load it into the Delta tables "customer" and "order".
2. The system should be able to remove "Null"/null and duplicate records from both tables.
3. The system should be able to create the "ordersummary" table if it does not exist and load the joined data into an SCD Type 2 table.
4. The system should be able to update the SCD Type 2 "ordersummary" table whenever there is a change in the "customer" table.
5. The system should be able to create the "customeraggregatespend" table if it does not exist and load the aggregated data into it.

- Non-Functional Requirements:**

1. The system should ensure data consistency and integrity throughout the data integration process.
2. The system should be able to handle large volumes of data efficiently.
3. The system should be designed to accommodate future changes in the data schema or business requirements.

- Assumptions and Dependencies:**

1. The CSV files are in the correct format and are located at the specified paths.
2. The Delta tables "customer" and "order" are created with the correct schema.
3. The catalog "gen_ai_poc_databrickscoe" and schema "sdic_wizard" exist in the database.

- Success Criteria:**

1. The data is successfully ingested from the CSV files into the Delta tables "customer" and "order".
2. The "ordersummary" table is created and populated with the joined data.
3. The SCD Type 2 "ordersummary" table is updated correctly whenever there is a change in the "customer" table.
4. The "customeraggregatespend" table is created and populated with the aggregated data.

- Acceptance Criteria:**

1. Verify that the data is correctly ingested from the CSV files into the Delta tables "customer" and "order".
2. Verify that the "ordersummary" table is created and populated with the joined data.

3. Verify that the SCD Type 2 "ordersummary" table is updated correctly whenever there is a change in the "customer" table.

4. Verify that the "customeraggregatespend" table is created and populated with the aggregated data.

By following this BRD, the development team should be able to design and implement a data integration and reporting solution that meets the business requirements and provides a robust and scalable framework for future enhancements.